Conformal Perceptual Shielding for Adaptive Bitrate Streaming

Saeed Zarbi[a], Leili Farzinvash[a,*], Pedram Salehpour[a]

[a]*Department of Computer Engineering, Faculty of Electrical and Computer Engineering, University of Tabriz, Tabriz, Iran*

Abstract

Client-side Adaptive Bitrate (ABR) controllers select, for each video chunk, one rung of the bitrate ladder to request. That request can stall playback when throughput drops, or waste bandwidth when top rungs are perceptually saturated, harming Quality of Experience (QoE). Prior solutions for these challenges ignored perceptual saturation and offered no coverage guarantee. Here, coverage refers to the probability that the actual throughput of the next chunk is at least as high as the lower bound used to determine whether the download can be completed before the buffer is empty. We propose the Certified Perceptual Shield (CPS), a model-agnostic wrapper for heuristic, model-based, or learned ABR policies. CPS, which has three components, works as follows. First, knee-based bandwidth banking lowers even a feasible proposal to the smallest rung whose Video Multimethod Assessment Fusion (VMAF) score lies within a perceptual budget ε of the proposal. Second, an online split-conformal throughput lower bound supplies that coverage without a parametric traffic model. Third, a feasibility check admits only rungs whose download time under that bound fits the buffer margin; otherwise CPS emits the lowest rung. This yields a per-chunk probabilistic stall certificate, not a zero-stall promise. An optional predictive mode (CPS-P) widens ε only when the buffer is low or a short look-ahead forecasts a dip. On 204 paired synthetic 5G episodes, banking reduced bitrate by 3.8–4.5% and rebuffering by 8.9–9.3% versus a safety-only shield across four non-learned controllers, with mean VMAF within one point and empirical coverage 0.919 against a 0.90 target. CPS adapts across regimes: on capacity-rich 5G links with saturated ladders it banks bandwidth (3.8–4.5% bitrate saved), and on capacity-limited real broadband traces—where top rungs are rarely jointly feasible and saturated—it shifts control to conformal feasibility, cutting rebuffering by 98.7%.

Keywords: Adaptive Bitrate Streaming, Runtime Shielding, VMAF, Conformal Prediction, Certified Safety, Quality of Experience

1. Introduction

Client-side Adaptive Bitrate (ABR) policies target average Quality of Experience (QoE). Classical controllers—Buffer-Based Adaptation (BBA) [1], BOLA [2], RobustMPC [3], bitrate-greedy selection (always requesting the top rung), and Proximal Policy Optimization (PPO) [4, 5]—optimized session averages. Those averages can obscure per-chunk risk under volatile sub-6 GHz and mmWave 5G throughput [6–8]. A comfortable buffer creates the opposite problem: wasted bandwidth. Commercial rate ladders often saturate in Video Multimethod Assessment Fusion (VMAF) [9]: further bitrate increases at the top often provide only marginal perceptual gains. Requesting those rungs when the buffer is healthy consumes bandwidth that could cushion the next dip.

Recent work attacked pieces of this problem, but not the same intersection. Conformal-ABR [10] wrapped a throughput predictor with split-conformal intervals and selected representations under an explicit stall-risk budget. SafeSABR [11] paired risk-calibrated learning with a runtime capacity auditor on volatile broadband. Kairos [12] strengthened Model Predictive Control (MPC) with buffer-conditioned uncertainty, yet without a conformal certificate. On the learning side, Comyco [13] folded VMAF into the training objective, and COREL [14] cast ABR as a Constrained Markov Decision Process (CMDP) with a stall-time penalty during rollouts. An expected-return or CMDP objective still leaves a per-chunk gap: guarantees hold over the sampled trajectory distribution, so an unseen blockage can induce an unsafe proposal at a single chunk boundary [14, 15]. Runtime post-shields sit between the base policy and the player and can overwrite that proposal before download [16, 17]. The standard repair, however, decrements the representation index—the discrete ladder index of the proposed rung—until a buffer-feasibility test passes. That repair is content-blind: it ignores VMAF saturation near the top of the ladder, and prior shields often used a fixed throughput scale without a stated coverage guarantee. None of these lines combines all three ingredients that we need: a model-agnostic post-shield that overwrites an unsafe proposal, a stated conformal throughput lower bound, and VMAF-knee bandwidth banking that downshifts early on saturated upper rungs to reserve headroom.

We introduce a *Certified Perceptual Shield* (CPS) that targets exactly this gap. Until a short warm-up of recent chunks fills the calibration window, CPS uses a fixed fallback throughput scale. It then switches to an online split-conformal lower bound with finite-sample coverage:

1. **VMAF-knee bandwidth banking.** Even a feasible action is lowered to the smallest rung whose VMAF lies within a small perceptual budget of the proposal. The downshift is

*Corresponding author.
Email address: l.farzinvash@tabrizu.ac.ir (Leili Farzinvash)

perceptually bounded, and reclaimed bytes bank as buffer for later chunks.

2. **Conformal throughput lower bound.** Download-time tests use an online split-conformal lower bound on throughput. Under residual exchangeability, the bound covers the next chunk with high probability at a chosen tolerated miss rate. Calibration uses recent residual ratios [18, 19].
3. **Certified feasibility.** After banking, CPS prefers a rung whose predicted download time under the throughput lower bound still fits in the remaining buffer (minus a small safety margin). Two emergency cases force the lowest rung instead: (i) no higher rung passes that test, or (ii) the buffer is already at or below a critical floor, so the player must request the cheapest chunk. Only when the downloaded rung passes the test and the buffer retains a minimal playout margin does this yield a per-chunk probabilistic stall certificate; it is not a promise of zero stalls.

A pure VMAF-ranking shield is not enough on a typical ladder. If quality never drops when the rung index rises, then among the feasible rungs the highest VMAF is always the highest index. Allowing a small perceptual loss does not force a downshift in that case, because the top feasible rung still wins the ranking. CPS therefore banks to the *smallest* rung inside the perceptual budget, rather than maximizing VMAF among feasible rungs. The same CPS rule wraps bitrate-greedy, BBA, and PPO hosts without changing their internal logic.

To separate banking from safety, we run three paired arms on identical episode seeds: RAW (unshielded), SAFETY (conformal feasibility only), and CERTIFIED (feasibility plus banking). The banking effect is the CERTIFIED minus SAFETY contrast. The safety effect compares SAFETY or CERTIFIED against RAW.

The main contributions are:

- **Certified Perceptual Shield.** A model-agnostic runtime projection that combines VMAF-knee bandwidth banking, a conformal throughput lower bound with finite-sample coverage under residual exchangeability, and certified buffer feasibility.
- **Isolated banking evaluation.** Paired RAW, SAFETY, and CERTIFIED rollouts on greedy, BBA, BOLA, RobustMPC, and PPO hosts. We report Wilcoxon tests on bitrate and rebuffering, and Two One-Sided Tests (TOST) of VMAF within a small perceptual tolerance [20].
- **Optional predictive banking (CPS-P).** This mode widens the perceptual budget dynamically from the base ε toward a risk budget $\varepsilon_{\text{risk}}$ when the buffer is low or a short look-ahead forecasts a dip.
- **Regime characterization.** Banking gains concentrate where ladders saturate at the top and hosts request those rungs (primary 5G suite). Real-trace broadband stress is dominated by safety projection.
- **Policy–shield co-design.** An optional PPO study with CPS in the training loop. We report the QoE stall weight at which this co-trained policy crosses a shield-at-eval-only baseline—trained without CPS, then shielded only at test time.

Quantitative results appear in Section 5. That section reports the paired 5G and broadband campaigns, optional CPS-P, and the co-design study.

The remainder of this paper is organized as follows. Section 2 reviews related work, and Section 3 formalizes the streaming model. Section 4 presents CPS and optional co-training. Section 5 reports the evaluation, and Section 6 concludes.

2. Related work

Three areas bear on client-side representation selection: model-based ABR, learned ABR, and constrained control with runtime shielding. Our focus is the control policy and its runtime action projection; throughput prediction lies outside this scope. Predictors such as CS2P [21] played a complementary role: they fed inputs to model-based or learned controllers [3, 4]. Beyond prediction, ABR surveys [8, 22–24] and 5G/mmWave measurements [6, 25] agree on one empirical lesson: control must remain robust when throughput fluctuates sharply.

2.1. Heuristic and model-based ABR

MPEG Dynamic Adaptive Streaming over HTTP (DASH) separates packaging from transport [26]. In this method, clients execute local adaptation policies based on ladder metadata and observed download conditions. Buffer-threshold heuristics such as BBA [1] remained simple and interpretable. BOLA [2], by contrast, derived its decisions from a Lyapunov-based utility formulation. RobustMPC [3] planned ahead. It selected representations by solving a short-horizon model predictive control problem under estimated future throughput. The payoff is interpretable thresholds and horizons. The cost is calibration fragility: performance degrades when hand-tuned thresholds, planning horizons, or the throughput forecasts that feed them fail to transfer across networks and devices [6, 7]. Moreover, VMAF scores often saturate on the upper rungs of a ladder, so those rungs add little perceptual quality [9]. Conventional feasibility rules still admit those upper rungs whenever the buffer test passes; they do not downshift early on saturated rungs to reserve throughput headroom. Other related efforts target different stages of the delivery pipeline. SODA [27] pursued smoothness-aware online control. Edge assistance [28] and Oboe-style automatic tuning [29] addressed other parts of the pipeline. Recent client-side designs also treat throughput uncertainty explicitly rather than relying on a single point forecast. Conformal-ABR [10] wrapped a throughput predictor with split-conformal intervals. It selected a representation under an explicit stall-risk budget. This yields distribution-free control of prediction error when the traces are exchangeable. SafeSABR [11] targeted volatile satellite broadband with risk-calibrated reinforcement learning. A runtime safety auditor then checked each policy proposal against conservative capacity bounds before download. Kairos [12] instead strengthened MPC-based ABR with streaming-aware prediction and buffer-conditioned uncertainty, improving robustness without a conformal certificate. These lines of work address

complementary failure modes: risk-aware control, runtime auditing, and robust planning. As positioned in Section 1, none of them combines a model-agnostic post-shield, a stated conformal throughput lower bound, and VMAF-knee banking; CPS targets that intersection at every chunk.

2.2. Deep reinforcement learning for ABR

Pensieve [4] established the template: a trace-driven Deep Reinforcement Learning (DRL) formulation for ABR. D-DASH [30] and more recent PPO-based controllers [31] followed this line. Their neural policies combine throughput histories, buffer state, and manifest features [32]. Comyco [13] trained by Imitation Learning (IL) against a VMAF-based QoE objective and showed that perceptual signals helped inside the learned policy. Loki [33] combined learned features with a rule-based model and reduced long-tail failures in production. Its rule module limited extreme decisions, but it provided no formal feasibility certificate. Fugu [34] deployed learned control in live networks. It exposed the distribution shift between simulation and deployment. That shift motivates runtime safeguards beyond expected-return training. Taken together, these systems motivate constrained learning and decision-time enforcement. As noted in Section 1, a Constrained Markov Decision Process (CMDP) still leaves a per-chunk enforcement gap that training alone cannot close.

2.3. Safe/constrained Reinforcement Learning (RL), shields, and the content-blindness gap

CMDPs and Lagrangian primal–dual methods optimize expected return under explicit constraints [35–37]. Safe-RL surveys catalogued related families, including shielding, Lyapunov, and risk-sensitive schemes [15]; Lyapunov-based safe RL [38] is a representative example. COREL [14] was closest to our training formulation. It cast ABR over commercial 5G/mmWave traces as a CMDP with a stall-time constraint. The CMDP was solved by a primal–dual actor–critic. That constraint, however, operated within training. It was not a standalone runtime projection that logs each intervention at decision time. Runtime shielding takes the opposite approach: it repairs a proposed action before execution and can log each intervention [16]. Surveys split the family into *pre-shielding*, which prunes the action set, and *post-shielding*, which overwrites an unsafe proposal [17]. By that taxonomy, CPS is a post-shield. Its safety set is drawn by a conformal feasibility test rather than a temporal-logic model.

Recent work has broadened both the learned and the uncertainty-aware branches of ABR. NetLLM [39] adapted a pre-trained Large Language Model (LLM) to bitrate selection. LingXi [40] tuned ABR objectives online from user-level engagement in a large production deployment. AAR [41] maximized worst-case QoE under bandwidth uncertainty with an MPC controller for low-latency live streaming. These designs improved expected or worst-case QoE, but none of them attached a per-chunk feasibility certificate to the selected representation. On the statistical side, conformal risk control [42] extended split-conformal guarantees from coverage to monotone losses. Online conformal updates [43] retained long-run coverage on non-exchangeable sequences. Conformal prediction has also become a standard tool for runtime verification and safe control of learning-enabled systems [44]. Recent shielded RL even filters actions with conformal safety margins [45]. CPS transfers this runtime-assurance pattern to ABR and couples it with VMAF-knee banking.

Table 1: Positioning of client-side ABR methods. Codes: Y = yes, ≈ = optional/partial, Imp. = implicit, N = no. **Constr.:** stall budget enforced during training. **Guard:** auditable action projection before download [16]. **Conf.:** conformal throughput lower bound with stated coverage [18]. **Bank:** VMAF-knee bandwidth banking within a perceptual budget ε.

Work	Year	Paradigm	Constr.	Guard	Conf.	Bank
BBA [1]	2014	Heuristic	N	N	N	N
BOLA [2]	2020	Lyapunov util.	Imp.	N	N	N
RobustMPC [3]	2015	MPC/lookahead	N	N	N	N
Pensieve [4]	2017	DRL	N	N	N	N
Comyco [13]	2019	Quality-aware IL	N	N	N	N
Loki [33]	2021	Rule–RL fusion	N	≈	N	N
COREL [14]	2023	Constr. DRL	Y	N	N	N
SODA [27]	2024	Online control	N	N	N	N
Conformal-ABR [10]	2026	Risk-controlled ABR	N	Y	Y	N
SafeSABR [11]	2026	Risk-cal. RL + auditor	Y	Y	≈	N
Kairos [12]	2025	MPC + uncertainty	N	N	N	N
NetLLM [39]	2024	LLM-adapted ABR	N	N	N	N
AAR [41]	2025	Robust max–min MPC	N	N	N	N
Ours (CPS)	—	Shield (±RL)	≈	**Y**	**Y**	**Y**

By that separation, CPS pairs an optional learned performance layer with a runtime post-shield. Learned policies such as Lagrangian PPO can pursue expected stall and smoothness budgets across training rollouts [14, 36], but they cannot certify per-chunk feasibility under conformal throughput bounds. Unlike Comyco, where VMAF enters only the learned objective [13], CPS reads the ladder at decision time to bank bandwidth proactively, rather than only repairing the representation index in an emergency. Table 1 compares representative client-side methods along these axes.

3. Model and problem formulation

This section formalizes client-side ABR at the chunk timescale. We assume a single client, exogenous trace throughput C_k^{tr} at chunk index k, and deterministic buffer updates given $(s_k, a_k^{\mathrm{exec}}, C_k^{\mathrm{tr}})$, where s_k is the client state and a_k^{exec} the executed ladder rung (Table 2). Here, deterministic means that the same inputs always yield the same next buffer and the same executed rung. Cross-layer scheduling and transport adaptation are out of scope. Certification of the conformal throughput lower bound is probabilistic (Section 4). Table 2 collects the symbols used throughout.

3.1. Streaming architecture and control loop

We work in the standard Hypertext Transfer Protocol (HTTP) adaptive-streaming abstraction [3, 4, 26]. A video origin or

Table 2: Summary of notation.

Symbol	Meaning
k, T_c, N_{ch}	chunk index, chunk duration, chunks per episode
L, $\mathcal{B}$={0, …, L−1}	number of ladder rungs, rung index set
a_k^{raw}, a_k^{exec}, ι_k	proposed rung, executed rung, intervention bit
B_k, m, B_{crit}	buffer level, safety margin, critical buffer
$V_k(j)$, $\sigma_k(j)$	VMAF and size (kb) of rung j at chunk k
ε, ε_{eff}, ε_{risk}	perceptual budget, effective/risk-mode budget
C_k^{tr}, $\hat{C}_k$, C_k^{act}, $\underline{C}_k$	trace rate C_k^{tr}; predictor $\hat{C}_k$; realized C_k^{act}; conformal lower bound $\underline{C}_k$ (kb/s)
W_{calib}, ρ_{fb}	residual warm-up length, fallback scale before calibration
α, 1−α	conformal miscoverage, target coverage
r=$C^{act}/\hat{C}$, W, κ	residual ratio, residual window, harmonic-mean window
$d_k(j)$, H, ε_0	download time of rung j, look-ahead, numerical guard
μ, β	logged-QoE smoothness and stall weights (Equation (2))
w, $w^\star$	QoE stall weight (sweep), mean-QoE crossover weight

Content Delivery Network (CDN) publishes L encoded rungs. We index them by $\mathcal{B} = \{0, \ldots, L-1\}$; the DASH or HTTP Live Streaming (HLS) manifest carries the matching per-chunk sizes and perceptual metadata. At decision step k, the client fetches chunk k while holding a playout buffer B_k. Decisions advance on the chunk clock, one step per playback duration T_c [3, 4]. The effective trace throughput C_k^{tr} is exogenous to the ABR client: the client sees only measured delivery rates, not how the base station schedules radio resources, as on mobile links where blockage and mobility abruptly change throughput [6, 7].

A base policy π (heuristic or learned) maps an augmented state s_k, including buffer occupancy, throughput history, manifest lookahead, and the VMAF ladder (defined below), to a proposed action $a_k^{raw} \in \mathcal{B}$. Before chunk k is downloaded, the Certified Perceptual Shield $\mathcal{S}$ applies the projection

$$(a_k^{exec}, \iota_k) = \mathcal{S}\left(s_k, a_k^{raw}\right), \qquad a_k^{exec} \in \mathcal{B}, \;\; \iota_k \in \{0, 1\}, \tag{1}$$

The projection returns the executed rung index a_k^{exec} and an intervention bit ι_k, which indicates whether the projection altered the proposal. Only a_k^{exec} feeds the chunk download and buffer dynamics Equation (4). Fig. 1 traces the $a_k^{raw} \to \mathcal{S} \to a_k^{exec}$ path. That path does not require a learned host: the actor, critic, and dual variables in the figure are an optional RL training branch and are unused at inference.

Optional training-time CMDP learning enforces budgets on stall time and smoothness in expectation over rollouts [36, 37]. It does not guarantee per-step stall avoidance from throughput uncertainty alone. At runtime, CPS projects a_k^{raw} to a feasible rung using a conformal lower bound $\underline{C}_k$; marginal coverage $C_k^{act} \geq \underline{C}_k$ holds at rate 1−α [18, 19].

3.2. State, action, reward, and nonlinear perceptual mapping

State s_k. The observation holds buffer occupancy, past throughput samples, and the previous decision. It also includes manifest lookahead: upcoming chunk sizes across the rung ladder, plus the spatial complexity index (SI) and temporal complexity index (TI), which measure texture and motion richness [46]. Higher SI/TI indicate richer spatial or temporal structure in the chunk. The VMAF ladder $\{V_k(j)\}_{j\in\mathcal{B}}$ completes the observation, where $V_k(j)$ is the perceptual score for chunk k at rung j [9]. This last component is essential: without $\{V_k(j)\}$, neither a learned host nor CPS can detect the content-dependent ladder saturation that bandwidth banking exploits.

Nonlinear, content-dependent fidelity. Saturation arises because quality does not scale linearly with rate: for each chunk k, the map $j \mapsto V_k(j)$ is not affine in the discrete rung index j. This map also varies across chunks. Different Group-of-Pictures (GOP) structures and texture content yield different marginal VMAF gains between adjacent rungs. Therefore, the bitrate-to-VMAF relationship is both nonlinear and content-dependent. Consequently, adjacent rungs may differ by a few VMAF points or by more than ten. Quality often saturates at the upper rungs of the ladder—precisely the structure that bandwidth banking exploits. CPS exploits any saturation that remains in a deployed ladder at decision time. In contrast, perceptually aware Just-Noticeable Difference (JND)-based per-title encoding aims to remove this saturation at authoring time [47].

Action. $a_k^{raw} \in \mathcal{B}$ is the policy's proposed rung index for chunk k; the environment executes only a_k^{exec}.

QoE surrogate. Keeping to RobustMPC/Pensieve reporting conventions [3, 4], we log a scalar per-chunk surrogate with fidelity $q_k \equiv V_k\left(a_k^{exec}\right)$:

$$\mathrm{QoE}_k^{(rep)} = q_k - \mu\,\mathrm{smooth}_k - \beta\,\tau_k^{stall}, \tag{2}$$

where $\mathrm{smooth}_k = |q_k - q_{k-1}|$ is the absolute VMAF change between adjacent chunks and τ_k^{stall} is the stall duration at step k. The first chunk takes a zero smoothness term. The weights μ and β are fixed in Section 5.1. The stall weight β is inherited from Pensieve's linear QoE [4], where it was set for bitrate in Mbps; we keep it unchanged rather than re-tuning it for the 0–100 VMAF scale, so the logged scalar weights stalls lightly. We therefore report rebuffering separately, and the co-design analysis sweeps the stall weight explicitly (Section 5.4). Session QoE sums over chunks: $\sum_{k\in\mathcal{K}_e} \mathrm{QoE}_k^{(rep)}$. For a learned host, optional RL training substitutes the Lagrangian CMDP objective in Section 3.5 [35, 36].

3.3. Ladder saturation

The evaluated system reports one VMAF value per rung and video (pooled per-frame scores), so shield decisions operate on a per-title session-mean ladder that does not vary across chunks. We keep the general notation $V_k(j)$ because the per-chunk ablation (Section 5.5) uses actual per-chunk ladders; in the headline runs $V_k(j) \equiv V(j)$. Table 3 summarizes session-mean ladders for 12 publicly available reference titles (five Blender animations and seven natural-motion Xiph DERF clips, including Ducks Take Off). Fig. 2 illustrates per-chunk spread on three titles with frame-level VMAF logs (Big Buck Bunny, Tears of Steel, and Sintel). Across the twelve-title suite, 67% exhibit a non-monotone session-mean ladder. Mid-ladder inversions concentrate at the 750→1200 kbps step; the top three rungs (1850, 2850, 6000 kbps) remain nondecreasing on every title. The mean top-rung gain (6000 vs. 2850 kbps) is only 5.61 VMAF

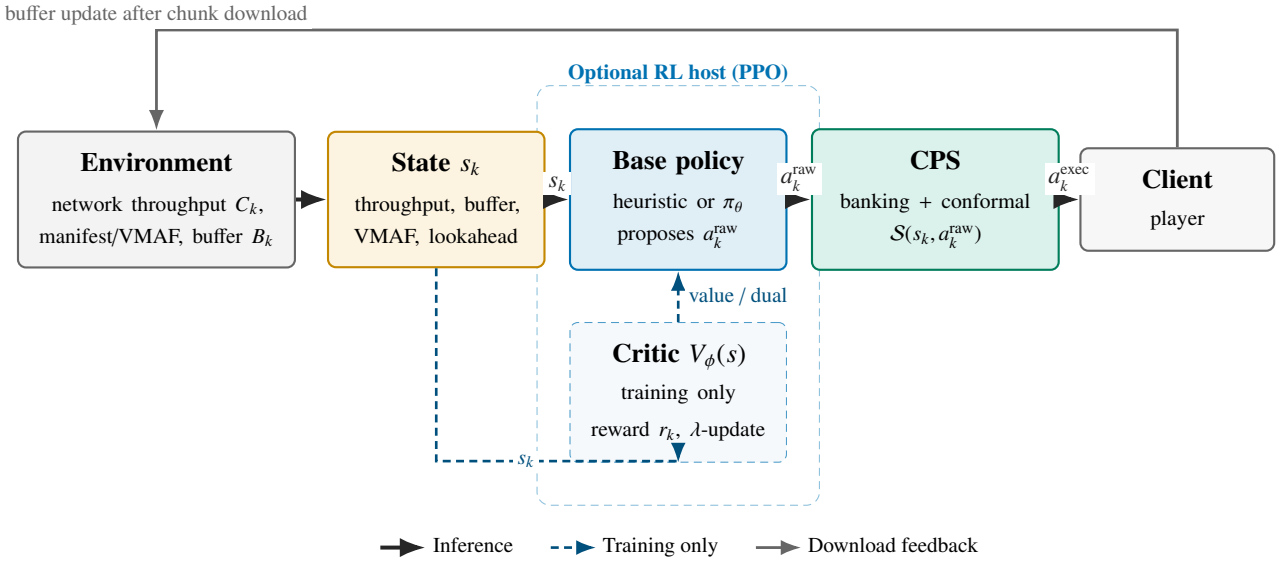


Figure 1: CPS control loop. Solid arrows: runtime inference path, where CPS projects the proposal a_k^{raw} to a_k^{exec} using $\underline{C}_k$. Dashed: optional RL training branch, unused at inference.

points. The session-mean top-rung gain averages across titles; saturated animation titles can appear nearly flat in Fig. 2 despite this mean. This pattern reflects strong saturation on animation titles (e.g. Big Buck Bunny at +0.03) but steeper climbs on natural content (e.g. Park Joy at +14.3). Two properties make banking worthwhile. First, many adjacent pairs differ by at most one VMAF point on saturated titles, so a single-step downgrade costs almost nothing perceptually. Second, the top rungs saturate frequently, allowing the knee rule in Equation (10) to bank bytes with no material quality loss. On the four titles with a nondecreasing session-mean ladder, max-VMAF and max-index feasible choices coincide (Proposition 2). The same equivalence holds for the upper three rungs on every title; greedy banking operates mainly there. CPS instead picks the smallest rung within the perceptual budget, preserving near-lossless downshifts under saturation. Shield decisions use session-mean ladders, but per-chunk VMAF can invert locally (Fig. 2); Section 5.5 tests banking on these per-chunk ladders.

3.4. Chunk-timescale dynamics

3.4.1. Throughput and latency

Let $C_k^{\text{tr}} > 0$ be the trace-derived effective throughput for chunk k. Section 4 defines predictor $\hat{C}_k$ (harmonic mean of recent realizations); the conformal construction yields lower bound $\underline{C}_k$ fed to the shield. The realized throughput of the executed download is C_k^{act}; in our trace-driven simulator $C_k^{\text{act}} = C_k^{\text{tr}}$. Given a_k^{exec}, the client downloads the manifest payload size $\sigma_k(a_k^{\text{exec}})$ in kilobits. Let $\tilde{C}_k = \max\{C_k^{\text{act}}, \varepsilon_0\}$ for a small constant $\varepsilon_0 > 0$ (the same numerical guard as in Equation (8)), and $d_k(j) = \sigma_k(j)/\tilde{C}_k$ for the nominal download duration of rung j. The executed chunk uses $d_k \equiv d_k(a_k^{\text{exec}})$ [3]. Feasibility timing under $\underline{C}_k$ is defined in Section 4.

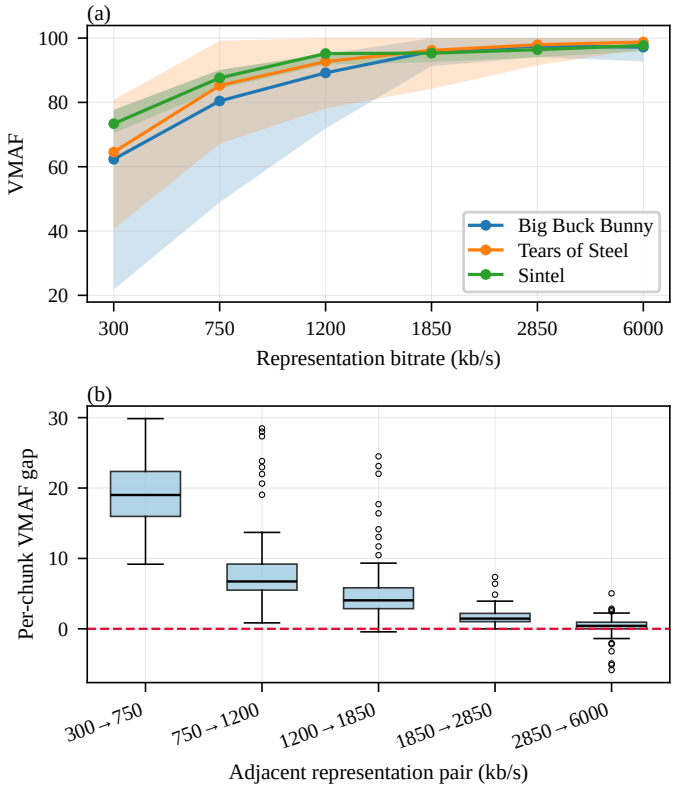


Figure 2: Per-chunk VMAF ladders for Big Buck Bunny, Tears of Steel, and Sintel (171 chunks). (a) Session-mean ladder (markers and lines); the shaded band of the same colour shows each title's per-chunk min–max range (blue: Big Buck Bunny, orange: Tears of Steel, green: Sintel). (b) Adjacent-rung VMAF gaps; dashed line: zero gain.

3.4.2. Buffer and stall

Let $B_k \geq 0$ be the buffered playback time when the download of chunk k begins [3, 4]. Rebuffering (stall) and the buffer update follow

$$\tau_k^{\text{stall}} = \max\{0,\ d_k - B_k\}, \tag{3}$$

Table 3: Session-mean representation ladder across twelve evaluation titles. Top gain: session-mean VMAF at 6000 kbps minus session-mean VMAF at 2850 kbps. Span: session-mean VMAF at 6000 kbps minus session-mean VMAF at 300 kbps. Mono.: non-decreasing session-mean ladder across rungs. Held-out (✓): excluded from PPO training. SI/TI: spatial/temporal complexity indices.

Video	Held-out	SI	TI	Top gain	Span	Mono.
Big Buck Bunny	—	60.6	47.1	+0.03	30.6	Yes
Tears of Steel	—	41.0	31.8	+4.09	26.1	No
Sintel	✓	50.4	14.7	+1.30	22.9	Yes
Elephants Dream	—	51.8	27.0	+7.87	31.2	No
Ducks Take Off	—	94.9	16.6	+11.59	36.6	No
Cosmos Laundromat	✓	42.6	38.3	+5.23	34.6	No
Park Joy	—	119.7	34.8	+14.31	46.5	No
Old Town Cross	—	61.2	12.6	+6.85	49.5	Yes
Rush Hour	—	27.7	11.7	+4.00	29.8	No
Into Tree	—	47.4	14.4	+9.12	36.1	No
Sunflower	—	38.0	16.5	+1.42	24.9	No
Kristen & Sara	✓	85.8	7.0	+1.53	37.6	Yes

$$B_{k+1} = \min\{\max\{0, B_k - d_k\} + T_c,\ B_{\max}\}, \tag{4}$$

where $B_{\max}$ is the player's buffer cap (12 s in our simulator unless noted). Randomness across episodes has a single source: trace and initial-condition sampling [22, 48, 49].

3.5. Optional Lagrangian training for RL hosts

When the base policy is learned, PPO training uses a Lagrangian-relaxed CMDP. The CMDP maximizes expected fidelity under budgets on trajectory-level rebuffering and smoothness [35–37]. An unconstrained Markov Decision Process (MDP) optimizes only $\mathbb{E}[\sum_k \gamma^k r_k]$. It places no explicit budget on stalls or smoothness. Set $c_k^{\text{stall}} \equiv \tau_k^{\text{stall}}$ and $c_k^{\text{sm}} \equiv \text{smooth}_k$. The undiscounted episodic rebuffering ratio is

$$\text{RB}(e) = \frac{1}{T_e^{\text{play}}} \sum_{k \in \mathcal{K}_e} c_k^{\text{stall}}. \tag{5}$$

The mean smoothness cost is

$$\text{SM}(e) = \frac{1}{N_{\text{ch}}} \sum_{k \in \mathcal{K}_e} c_k^{\text{sm}}, \tag{6}$$

with targets δ_{rf}=0.05 (rebuffering-ratio budget) and δ_{sm}=3.5 (mean smoothness budget in VMAF points) [3]. Per-step normalized costs $\widetilde{\text{RB}}_k$ and $\widetilde{\text{SM}}_k$ yield the Lagrangian-relaxed PPO reward

$$r_k^{\text{train}} = \frac{1}{100}\Big(q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k - w_b\, \phi(B_k) - w_\ell\, \psi_k^{\text{Lya}}\Big), \tag{7}$$

where projected ascent on the empirical constraint gaps updates (λ_r, λ_s) [36]. The buffer-deviation term $\phi(B_k)$ and the Lyapunov drift penalty ψ_k^{Lya} are optional [50]. They steady the optimization. They do not certify per-chunk feasibility. Finite-sample PPO with a clipped objective cannot provide that certificate [5]. CPS supplies it at deployment time. The implementation uses Stable-Baselines 3 PPO [5, 51]: two Tanh Multi-Layer Perceptrons (MLPs) of width 256, Generalized Advantage Estimation (GAE) with discount γ=0.99 and λ_{GAE}=0.95, dual step sizes η_r=0.012 and η_s=0.003, and projected ranges $\lambda_r \in [4.3, 40]$, $\lambda_s \in [0.3, 2.5]$ after a 60-episode warm-up. The discount γ and the GAE parameter λ_{GAE} are the standard PPO settings of Stable-Baselines 3 and GAE [5, 51]. In contrast, λ_r and λ_s are Lagrangian dual multipliers rather than fixed PPO hyperparameters; after the warm-up, projected ascent on the constraint gaps updates them within the ranges above. These ranges follow from the VMAF scale: the lower bound of λ_r lies close to the stall weight used in evaluation, so the penalty is effective from the start, and the upper bound keeps the rebuffering cost commensurate with fidelity units without overwhelming the reward. The step sizes η_r and η_s are the dual ascent rates; for constraint gaps on the order of a few hundredths, they yield gentle but stable updates, and η_s is smaller because the smoothness constraint acts on the per-point scale. Both rates were stabilized by monitoring the dual traces and the achieved-versus-target constraint gap during training. Co-design wraps the training environment in CPS, so the actor–critic observes the projected action distribution. Shield-at-eval-only policies train without the shield.

4. Method: Certified Perceptual Shield

CPS is a model-agnostic wrapper around any base policy that proposes $a_k^{\text{raw}} \in \mathcal{B}$. A base policy proposes a ladder rung; CPS applies a projection before download; the player then updates the buffer under realized throughput. The projection is deterministic: for fixed inputs it always returns the same executed rung. What it certifies is probabilistic: feasibility is tested against a conformal throughput lower bound with a stated coverage guarantee. The rest of this section covers the conformal estimator, the banking and feasibility projection, a predictive extension, optional PPO co-training, and the evaluation arms.

4.1. Conformal throughput lower bound

The point predictor $\hat{C}_k$ is the harmonic mean of the last κ observed throughputs, following the RobustMPC throughput estimator [3]. After each step, the multiplicative residual $r = C^{\text{act}}/\hat{C}$ is appended to a rolling window of size W. With n residuals in that window, split conformal picks a conservative residual quantile at the target miss rate α: the entry at rank $\lfloor \alpha(n+1) \rfloor$ in the sorted list (1-based indexing). The factor $n+1$ is the standard finite-sample correction; a linearly interpolated empirical quantile would under-cover on short episodes [18, 19]. If that rank is undefined—or while $n < W_{\text{calib}}$ with W_{calib}=20—we fall back to a fixed scale ρ_{fb}=0.80. That warm-up is not the finite-sample certificate. The lower bound reads $\underline{C}_k = \hat{C}_k \cdot q_\alpha$, clipped to stay at or below $\hat{C}_k$. Under residual exchangeability, the construction guarantees $\mathbb{P}(C^{\text{act}} \geq \underline{C}_k) \geq 1 - \alpha$ once the residual window has length at least W_{calib}. Given $\underline{C}_k$, the certified download duration of rung j is

$$d_k(j) = \min\left\{\frac{\sigma_k(j)}{\underline{C}_k + \varepsilon_0},\ 60\right\}. \tag{8}$$

Both $\sigma_k(j)$ and $\underline{C}_k$ use kilobit units (kb and kb/s), so their ratio is already in seconds; ε_0>0 and the outer $\min\{\cdot, 60\}$ are numerical guards against near-zero denominators.

Proposition 1 (**Per-chunk stall certificate**). *Assume chunk k, executed index j, and conformal lower bound $\underline{C}_k$ computed with miscoverage α. Suppose the following three conditions hold:*

(i) $B_k \geq 0.1$ s *(buffer floor);*
(ii) $\bar{d}_k(j) \leq \max\{B_k - m, 0.1\}$, *where* $\bar{d}_k(j) = \sigma_k(j)/\underline{C}_k$ *is the unguarded certified duration;*
(iii) $C_k^{\text{act}} \geq \underline{C}_k$.

Then $\tau_k^{\text{stall}} = 0$. *The shield (Section 4.2) tests the guarded duration* $d_k(j)$ *of Equation* (8)*, which equals* $\bar{d}_k(j) \cdot \underline{C}_k/(\underline{C}_k + \varepsilon_0)$ *whenever the* 60 s *cap is inactive; with* $\varepsilon_0 \ll \underline{C}_k$ *the two tests coincide up to this negligible factor, and the proposition applies to chunks that pass (ii). Condition (i) is required so that* $\max\{B_k - m, 0.1\} \leq B_k$. *Separately, under residual exchangeability,* $\mathbb{P}(C_k^{\text{act}} \geq \underline{C}_k) \geq 1 - \alpha$ *at each step where the finite-sample bound is defined* ($n \geq W_{\text{calib}}$)*. That coverage statement concerns the throughput lower bound alone; it is not itself a stall claim.*

Proof sketch. By (iii) and $\underline{C}_k > 0$, $\tilde{C}_k \geq C_k^{\text{act}} \geq \underline{C}_k$, so the realized download time satisfies $\sigma_k(j)/\tilde{C}_k \leq \bar{d}_k(j)$. By (ii) and (i), $\bar{d}_k(j) \leq \max\{B_k - m, 0.1\} \leq B_k$; hence $\tau_k^{\text{stall}} = \max\{0, d_k - B_k\} = 0$. Coverage follows from standard split-conformal prediction with the $(n+1)$ rank correction [18, 19]. □

Scope of the guarantee. Two parts of the certificate behave differently. The feasibility guarantee is unconditional only when the executed rung satisfies $\bar{d}_k(j) \leq \max\{B_k - m, 0.1\}$ and $B_k \geq 0.1$ s. If the realized throughput then meets or exceeds $\underline{C}_k$, that chunk cannot stall. The shield can still select rung 0 when $B_k \leq B_{\text{crit}}$, or when the decrement loop reaches 0 while $d_k(0)$ fails the test (Section 4.2). Those chunks are outside the certificate.

Only the throughput bound is probabilistic. It provides marginal coverage over the exchangeable residual window, $\mathbb{P}(C_k^{\text{act}} \geq \underline{C}_k) \geq 1 - \alpha$, averaged over residual draws rather than per chunk, and only after warm-up. It does not provide coverage conditional on the current buffer or throughput regime. Distribution-free split conformal cannot do so without stronger assumptions [19]. CPS therefore certifies feasibility on every chunk whose bound holds and whose rung passed the test—under exchangeability the bound fails with probability at most α; it does not promise a stall-free session.

Abrupt 5G/mmWave Line-of-Sight/Non-Line-of-Sight (LOS/NLOS) transitions and scheduling shifts can break residual exchangeability [6]. Our online defense is a sliding residual window of length W. The recent ratios play two roles: they calibrate q_α, and they are the ratios immediately preceding the chunk we predict. The bound therefore tracks local nonstationarity, even though finite-sample exactness holds only on an exchangeable window. If drift is large, coverage decays. The decay is bounded by a total-variation term between calibration and test residuals [52]. A weighted-conformal variant that down-weights stale residuals is a natural extension, which we leave to future work. Empirically, we treat reported coverage as the operational check of Proposition 1. The headline rate averages over every chunk with a defined bound, including the W_{calib} warm-up steps that use ρ_{fb}. On the primary 5G pool, that inclusive rate is 0.919 against the target 0.90. Restricting to post–warm-up chunks alone raises coverage to 0.931 (warm-up chunks alone: 0.903); Section 5 reports both views. Real broadband traces (Section 5.3) and tighter α (Section 5.6) can sit marginally below the target. This is what the theory predicts: short calibration windows and nonstationary traces both erode finite-sample coverage.

Algorithm 1 Certified Perceptual Shield (compact)

Input: a^{raw}, B_k, $\{\sigma_k(j), V_k(j)\}$, conformal estimator; parameters $(\varepsilon, m, B_{\text{crit}})$
Output: (a^{exec}, ι) with $\iota = \mathbf{1}[a^{\text{exec}} \neq a^{\text{raw}}]$
$a \leftarrow \text{clip}(a^{\text{raw}})$; $\underline{C}_k \leftarrow$ conformal lower bound; define $d_k(\cdot)$ via Equation (8)
if $B_k \leq B_{\text{crit}}$ **then**
    return $(0, \mathbf{1}[a \neq 0])$
end if
compute ε_{eff} (Section 4.3); $j \leftarrow j_k^{\text{knee}}(a)$ via Equation (10) with ε_{eff}
while $j > 0$ and $d_k(j) > \max\{B_k - m, 0.1\}$ **do**
    $j \leftarrow j - 1$
end while
$a^{\text{exec}} \leftarrow \min\{a, j\}$; **return** $(a^{\text{exec}}, \mathbf{1}[a^{\text{exec}} \neq a])$

4.2. Banking and certified feasibility

Safety set. Indices that never exceed the proposal and fit the buffer margin m form

$$\mathcal{A}_{\text{safe},k}(a) = \{j \leq a : \; d_k(j) \leq \max\{B_k - m, 0.1\}\}. \tag{9}$$

VMAF-knee banking. The perceptual budget ε is a tolerance in VMAF points. Even when a is feasible, CPS steps down to the smallest rung whose VMAF lies within ε of the proposal, namely

$$j_k^{\text{knee}}(a) = \min\{j \leq a : \; V_k(a) - V_k(j) \leq \varepsilon\}, \tag{10}$$

The shield then projects $j_k^{\text{knee}}(a)$ onto $\mathcal{A}_{\text{safe},k}(a)$; safety always takes precedence over ε.

Remark. Banking may run a lower rung than the proposal, and a lower rung can only shorten $d_k(\cdot)$. Once Equation (9) holds, safety projection therefore keeps the certificate intact.

Algorithm 1 states the CPS projection that realizes Equations (9)–(10). If $B_k \leq B_{\text{crit}}$, CPS selects the lowest rung with no feasibility test. Otherwise it computes the knee index from Equation (10) under the effective budget ε_{eff}. This budget equals ε unless predictive mode is on (Section 4.3). CPS then decrements the index until the certified duration in Equation (8) respects the margin m, or until the index is 0. If rung 0 still fails the test, the procedure still selects it so that playback can continue: it is an emergency best-effort choice without the stall certificate of Proposition 1. The same holds for the B_{crit} path. The shield never upgrades a proposal. Diagnostics log banked bits, VMAF surrendered, and whether realized throughput met $\underline{C}_k$ (empirical coverage).

Banking distinguishes CPS from prior VMAF-ranking shields, and the distinction is substantive. On any nondecreasing prefix of a ladder, a ranking shield provably banks nothing.

Proposition 2 (**Ranking shields collapse on monotone ladders**). *Assume chunk k and suppose the VMAF ladder is nondecreasing in the rung index,* $V_k(0) \leq \cdots \leq V_k(L-1)$*, and that download time* $d_k(\cdot)$ *is nondecreasing in the rung index.*

Let $\mathcal{F} = \{j : d_k(j) \le \max\{B_k - m, 0.1\}\}$ be the feasible set, as in Equation (9)*. Then any shield that selects* $\arg\max_{j\in\mathcal{F}} V_k(j)$ *selects the highest feasible index in $\mathcal{F}$, independently of any perceptual-loss budget ε.*

Proof. Because d_k is nondecreasing, $\mathcal{F}$ is a contiguous prefix of the ladder ending at its largest index. Because V_k is also nondecreasing, the feasible rung with highest VMAF is exactly that largest index; the budget ε never changes the choice. □

Ranking by retained VMAF is therefore equivalent to selecting the highest feasible index, and it frees no bandwidth. CPS instead uses the proposal's quality and steps down to the smallest rung within ε of it, as stated in Equation (10). This mechanism lets CPS exploit rate–quality saturation: when top rungs carry near-identical VMAF, the knee lies well below the highest feasible index, and the freed bytes bank as buffer.

The evaluation isolates banking from safety with three paired arms (Raw, Safety, Certified); Section 5.1 defines them relative to Algorithm 1.

4.3. Predictive and risk-aware perceptual budget

A fixed perceptual budget spends the same tolerance on every chunk. That spend is unnecessary while the buffer is comfortable, and it can be too small when a dip is imminent. CPS-P, the predictive mode introduced above, exists to separate those two cases: it keeps the base budget ε in the comfortable regime and widens it only under the risk triggers below. Section 5.3 compares this mode with a larger static budget.

The predictive budget widens when slack is scarce: it banks lightly when buffer headroom is ample and more aggressively when risk arises. While the buffer stays comfortable, $\varepsilon_{\text{eff}}{=}\varepsilon$ holds quality steady. Two risk triggers widen the budget: (i) the buffer B_k falls below a risk threshold B_{risk}; (ii) a short H-step look-ahead under a planning throughput floor (a low quantile of recent observations) forecasts a dip below B_{risk}. In either case, the effective budget is

$$\varepsilon_{\text{eff}} = \varepsilon + (\varepsilon_{\text{risk}} - \varepsilon) \cdot f(B_k, \text{lookahead}), \tag{11}$$

where $\varepsilon_{\text{risk}} > \varepsilon$ is the wider risk budget. Let $\underline{C}_k^{\text{plan}}$ be the planning throughput: the conformal lower bound, optionally floored by a low quantile of recent observations when dip forecasting is enabled. Simulate H future downloads of the proposed rung under $\underline{C}_k^{\text{plan}}$, write $\hat{B}_{k+h}$ for the resulting buffer after step h, and let $\hat{B}_k^{\min} = \min_{1\le h\le H} \hat{B}_{k+h}$. Then

$$f(B_k, \text{lookahead}) = \begin{cases} 1, & \hat{B}_k^{\min} < B_{\text{risk}}, \\ \text{clip}\Big(\dfrac{B_{\text{risk}} - B_k}{B_{\text{risk}}}, 0, 1\Big), & \text{otherwise}, \end{cases} \tag{12}$$

so f rises from 0 at a comfortable buffer to 1 at or below B_{risk}, and jumps to 1 if the look-ahead predicts a breach. The shield can therefore pre-bank more deeply, before a stall forces an emergency cut. CPS-P denotes the same CPS projection with this risk-aware ε_{eff} in place of a fixed ε; the conformal bound and feasibility test are unchanged. Operating values for ε, $\varepsilon_{\text{risk}}$, B_{risk}, and H appear in Section 5.1.

4.4. Optional learned hosts and co-design

CPS wraps any host and does not require a learned policy; heuristic and model-based controllers use the same projection. When we optionally study a learned host, we train a standard PPO policy (optionally Lagrangian-shaped) with the Stable-Baselines 3 library [5, 51] on content-aware observations: throughput history, buffer, spatial and temporal complexity (SI/TI), and the VMAF ladder. We do not propose a new RL algorithm; PPO is one optional host among others. The subsections above define a host-agnostic shield. This subsection asks a separate question: whether training inside that shield changes the policy the shield later wraps. By *co-design* we mean combining learning with the shield during training: wrapping the training environment in CPS changes the action distribution observed by the actor–critic, so the policy can learn to propose bolder rungs that banking later trims for safety. To gauge the effect, we compare two policies under identical 5G train/test provenance: one trained without the shield (shield-at-eval-only), one co-trained with CPS. Both face CPS-P at test time.

4.5. Implementation notes

Shield and predictive-banking defaults for the primary experiments appear in Section 5.1 (Table 4). For the first W_{calib} chunks of every episode, before the residual window supports a finite-sample quantile, the fallback scale is set to ρ_{fb}; that warm-up lies outside the certificate. Each episode has 48 chunks and the residual window resets at the episode boundary, so a long nominal window W need not fill within one episode. Empirical coverage is the fraction of steps with $C^{\text{act}} \ge \underline{C}_k$, counted over all chunks including warm-up. Per decision, the projection runs in $O(L{+}H)$ over the L ladder rungs and an H-step look-ahead. Conformal-quantile upkeep on the length-W residual window costs $O(\log W)$ per update with an ordered container (insertion, eviction, and rank lookup). Across 20,000 rollout decisions on a commodity CPU (L=6, full predictive shield), the per-decision wall-clock elapsed time is 131 μs at the median and 310 μs at the 99th percentile. This is roughly 10^{-5} of a 4 s chunk budget. CPS therefore runs inline, and the control loop is unaffected.

5. Experimental evaluation

We organize the evaluation around four research questions:

RQ1 *Banking vs. safety.* Does VMAF-knee banking reduce bitrate and rebuffering relative to conformal safety-only projection, with VMAF inside $\pm\varepsilon$?

RQ2 *Coverage.* Does empirical conformal coverage meet the finite-sample target $1{-}\alpha$?

RQ3 *Budget extensions.* Relative to the default budget ε, do CPS-P or a larger static ε change bitrate and rebuffering savings?

RQ4 *Robustness.* Does the banking stage (banked bitrate and realized per-chunk perceptual cost) remain effective on non-monotone per-chunk ladders? Do the RQ1–RQ2 results remain stable as ε and α vary?

A single-seed co-design study appears in Section 5.4.

5.1. Experimental setup and datasets

The testbed is a multi-video, chunk-timescale simulator with N_{ch}=48 chunks per episode, T_{c}=4 s, and buffer cap $B_{\max}$=12 s unless noted. It replays a throughput trace chunk by chunk: for each chunk it applies the executed rung's size, computes the download time and stall from the trace throughput, and updates the buffer with Equation (4). It drives twelve publicly available reference titles with session-mean VMAF ladders [9]: five Blender open animations [53]—Big Buck Bunny, Tears of Steel, Sintel, Elephants Dream, and Cosmos Laundromat—and seven natural-motion Xiph DERF clips—Park Joy, Old Town Cross, Rush Hour, Into Tree, Sunflower, Kristen & Sara, and Ducks Take Off [54]. We select them for ladder structure, not to cover many content genres: saturation-rich animation such as Big Buck Bunny and Tears of Steel, steep natural motion such as Park Joy and Ducks Take Off, wide-span DERF clips such as Into Tree and Old Town Cross, and mixed held-out titles such as Sintel, Cosmos Laundromat, and Kristen & Sara for generalization checks. Nine titles appear in PPO co-design training; three—Sintel, Cosmos Laundromat, and Kristen & Sara—are never seen during PPO training, to test generalization to unseen titles. Because CPS acts only on the observed VMAF ladder and buffer state, ladder shape—not semantic content—drives its effect. The twelve-title suite therefore spans both saturated and steep/inverted session-mean ladders, with 67% non-monotone.

Primary 5G suite. Public per-chunk 5G/mmWave logs suitable for chunk-level ABR replay remain scarce [6, 25]. We therefore built a synthetic 5G trace generator; its test traces, held out from the co-design training seeds, form the primary pool of 204 paired episodes. The generator is configured as follows. Each trace lasts 1000 s at a baseline of 9000 kbps. Every 40 s, starting at offset 20 + (i mod M), a 4 s interval is replaced by a dip drawn uniformly from [500, 1200] kbps on the test set and from [400, 1300] kbps on the training set. The trace is then scaled by an independent factor from U(0.85, 1.15) and clipped to [80, 50000] kbps. The test set contains 50 traces (seed 42, M=17). The training set contains 80 traces (seed 7000, M=23); here i indexes the dip and M is a trace-generator offset period, unrelated to the buffer margin m. The broadband suite below provides a real-trace counterpart. Train and test trace pools come from the same generator with different seeds and dip ranges, so they are seed-disjoint rather than source-disjoint. Co-design policies train on the training pool alone and face held-out test seeds.

Broadband suite. The broadband suite uses real-measured traces: the Federal Communications Commission (FCC) Measuring Broadband America (MBA) fixed-line logs [48] and the Norway/Nornet mobile-broadband mobility logs [49], with 27 held-out test traces that are source-disjoint from training. Both are widely used public datasets in ABR research. They provide a real-network counterpart to the primary 5G suite and show where banking offers limited headroom.

Hosts. Five host families cover the ABR design space: bitrate-greedy, always at the top rung; BBA; BOLA, the Lyapunov buffer-based rule of the dash.js reference player [2, 55]; robust model-predictive control via RobustMPC [3]; and two PPO policies: a content-aware policy that also observes SI/TI and the VMAF ladder (Section 4.4), and a content-blind Pensieve-style policy without these content features [4]. RobustMPC is a five-segment planner over a throughput prediction discounted by recent worst-case error. CPS carries no host bias: one wrapper, applied at inference across all five families, tuned to none of them.

Table 4: CPS default hyperparameters (primary 5G suite).

Parameter	Value
Perceptual budget ε	1.0 VMAF point
Conformal miscoverage α	0.10; target coverage 0.90
Residual window W / warm-up W_{calib}	200 / 20
Fallback scale ρ_{fb} / predictor κ	0.80 / 5
Buffer margin m / critical B_{crit}	0.5 s / 0.3 s
Risk budget $\varepsilon_{\text{risk}}$ / B_{risk}	4.0 / 8 s
Look-ahead horizon H / forecast quantile	6 / 0.2

W=200 is a nominal residual capacity; each episode has 48 chunks and resets the window, so at most 47 residuals are available within an episode. Look-ahead uses H=6 steps in all reported CPS-P runs.

Evaluation configurations and paired design. We call each shield configuration an *arm*. Every host rolls out Raw, Safety, and Certified on identical Python seeds—the same video–trace pairs. Raw runs a^{raw} unshielded. Safety runs Algorithm 1 with banking switched off by setting $j^{\text{knee}}=a$, so only conformal feasibility remains. Certified runs the full CPS (banking plus feasibility). Pairing fixes the random seed and the (video, trace) assignment across arms, so metrics differ only because of shielding, not because of resampling. In paired evaluations every method observes the same trace C_k^{tr}. Each paired contrast is the per-episode metric difference on the same (video, trace) seed across arms; for example, the Certified-arm value minus the Safety-arm value on the same episode.

Metrics and statistical tests. We report mean delivered bitrate, mean rebuffering per episode, stall-free rate, mean VMAF, intervention rate, and empirical conformal coverage. Significance rests on paired Wilcoxon signed-rank tests with zero-method `zsplit` and on TOST for equivalence of VMAF within $\pm\varepsilon$=1.0 [20]. Table 4 lists the primary-5G defaults, including α=0.10 with coverage target 0.90, ε=1.0, warm-up W_{calib}=20 with fallback ρ_{fb}=0.80, and the CPS-P risk settings. We set ε=1.0 as both the knee budget and the TOST margin.

Logged QoE (Equation (2)) uses μ=1.0 per absolute VMAF-point change and β=4.3 per stall second throughout the shared simulator; with β=4.3, one stall second costs only 4.3 VMAF points. One VMAF point is a small fraction of typical adjacent-rung and top-rung gaps on our ladders in Table 3 and is conservative relative to perceptually aware per-title encoding practice [47]. It is an engineering design choice aligned with Equation (10), not a preregistered clinical equivalence margin [20]. Because the knee rule and this TOST both use the session-mean ladder, alignment of the session-mean VMAF change with $\pm\varepsilon$ is expected; Section 5.5 therefore validates budget adherence independently on true per-chunk VMAF. The scalar in Equation (2) is only the logging definition. Only the co-design sweep uses the linear score QoE(w) defined in Section 5.4. The p-values

we quote for the primary arms are raw and two-sided Wilcoxon signed-rank tests with zero-method `zsplit`. When bitrate and rebuffering share the same signed-rank statistic on a host—for example, greedy certified–safety p=8.5×10^{-4} for both—the identity is mechanical, not a reporting error. The co-design values 5.0×10^{-5} and 5.0×10^{-5} equal 1/20,000, the floor of a 20,000-draw sign-flip test; they are not Wilcoxon tail probabilities. The bootstrap intervals in Section 5.4 are the magnitude statement for that contrast. For the eight primary certified–safety bitrate and rebuffering contrasts across four non-learned hosts, the largest raw p is 8.5×10^{-4}; all eight remain below the Holm–Bonferroni threshold at family-wise 0.05. One caveat concerns content diversity. Each episode is one (video, trace) pair: the video is drawn uniformly from the twelve titles and the trace from the held-out pool. Observations therefore cluster by title, and effective independent-content diversity is twelve, not 204. We treat the reported p-values as within-pair evidence on that paired design, not as evidence from 204 independent titles.

5.2. Model-agnostic banking on 5G (RQ1–RQ2)

Table 5 reports absolute metrics for all three arms—Raw, Safety, and Certified—on the primary 5G test traces for five host families: greedy, BBA, BOLA, RobustMPC, and a Pensieve-style negative control. The table lists the episode means and significance tests for every arm. Banking behaves consistently across all four non-learned hosts on the twelve-title suite. Relative to Safety, the Certified arm saves 4.5%, 4.1%, 4.1%, and 3.8% bitrate on greedy, BBA, BOLA, and RobustMPC, respectively, and reduces rebuffering by 9.3%, 8.9%, 8.9%, and 9.2% (greedy rebuffering Wilcoxon p=8.5×10^{-4}). Fig. 3 summarizes only the Certified-vs.-Safety contrast and omits Raw. On steep, non-saturated ladders the absolute savings stay modest. This is what the mechanism predicts: banking pays off only where the ladder saturates. Every host nevertheless remains perceptually near-lossless. Partitioning the greedy certified–safety contrast by session-mean top-rung VMAF gain makes that dependence explicit (Table 3). On titles with top gain at most 2 VMAF points—Big Buck Bunny, Sintel, Sunflower, and Kristen & Sara, n=60 episodes—banking saves 4.6% bitrate and 9.9% rebuffering. On steep titles with top gain at least 9—Park Joy, Ducks Take Off, and Into Tree, n=51—the same contrast shrinks to 2.1% bitrate and 3.7% rebuffering. The pooled greedy headline therefore mixes both regimes rather than reflecting a saturation-only subset, saving 4.5% bitrate and 9.3% rebuffering. Mean VMAF differences are perceptually negligible, at −0.01, 0.17, 0.18, and 0.02 points across greedy, BBA, BOLA, and RobustMPC, with every value inside $\pm\varepsilon$ by TOST.

Safety and banking contribute differently. On greedy, mean rebuffering per ∼192 s episode of 48 chunks falls from 85.5 s for Raw to 41.9 s for Safety and 38.0 s for Certified. Against Raw, certified greedy reduces rebuffering by 55.5%, versus 51.0% for BBA. Conformal safety removes most of the stall time relative to Raw; banking adds the certified–safety margin on top. Fig. 5 and Fig. 6 show per-episode operating points and rebuffering Cumulative Distribution Functions (CDFs) for greedy and BBA.

Coverage follows a distinct pattern on the primary 5G pool. The inclusive empirical coverage—averaging every chunk, including the W_{calib} fallback warm-up—is 0.919 for greedy and 0.919 for BBA against a 0.90 target. On the greedy certified arm, warm-up chunks alone cover at 0.903 and post–warm-up chunks at 0.931, so the headline value 0.919 is not a pure finite-sample certificate rate. Because coverage depends on the throughput trace and the conformal calibrator rather than the base policy, the inclusive rate equals 0.919 across every host and shield variant on the 5G test set; see Fig. 4. Proposition 1 is a per-chunk probabilistic certificate, not a per-episode stall-free one. On the greedy certified arm we attribute every second of residual rebuffering on the primary 5G pool of 204 paired episodes to an exclusive cause. When the residual window is shorter than the conformal warm-up of W_{calib}=20, 91.3% of that stall time occurs; the bound then uses the fixed fallback scale and is not the finite-sample certificate. After warm-up, 8.7% of residual stall time falls on chunks where realized throughput misses its conformal lower bound. A further 0.0% falls on chunks where the bound holds, matching Proposition 1. The $B_k \le B_{\text{crit}}$ emergency path contributes 0.0% of stall seconds on this suite. Safety still leaves mean rebuffering of 41.9 s on a ∼192 s greedy episode, against 85.5 s for Raw; most of the residual certified stall mass sits in the warm-up path above. Pensieve, the negative control, degrades to near-floor bitrates with near-zero stalls, so CPS rarely intervenes. Banking cannot rescue a policy that never requested saturated rungs.

Greedy proposes the top rung every time, so nearly every banking downshift or feasibility repair appears as an intervention; its intervention rate in Table 5 is therefore high. The rate gauges how far a host sits from the certified operating point, not how much of the outcome to attribute to the shield. This is why we run the model-agnostic contrast across hosts spanning the spectrum.

The certified intervention rate falls as the host becomes more conservative: 70.4% for greedy, 66.1% for BBA, 61.1% for BOLA, 31.1% for RobustMPC, and 7.4% for the Pensieve-style host. The sharper the base policy, the lighter the shield's load. Coverage nevertheless holds at 0.919 throughout. The certified metrics therefore depend increasingly on the base policy's own decisions, and the shield acts only where banking headroom remains. For RQ1 and RQ2 the message is simple: on every non-learned host, banking saves bitrate and cuts rebuffering with mean VMAF inside $\pm\varepsilon$. Inclusive coverage meets 0.90 on the primary 5G pool; the post–warm-up rate 0.931 is the closer operational check of the finite-sample certificate.

5.3. Predictive banking and broadband boundary (RQ3)

With the risk-aware budget ε_{eff} of Equation (11) and dip forecasting enabled under CPS-P, which starts from the default base budget ε=1.0 and widens it up to $\varepsilon_{\text{risk}}$=4 VMAF points, certified banking vs. safety reaches a 14.6% bitrate reduction for greedy and 12.5% for BBA, and cuts rebuffering by 12.6% and 13.0%, respectively, at mean VMAF changes of −0.23 and −0.02. Both contrasts are against the same Safety arm; with a fixed ε=1.0, the corresponding bitrate savings are 4.5% and 4.1% (Table 5). A static budget ε=2 on the same greedy host saves 14.4% bitrate and 30.8% rebuffering at a mean VMAF change of −0.17 (Table 10), compared with 14.6%, 12.6%, and −0.23 under CPS-P.

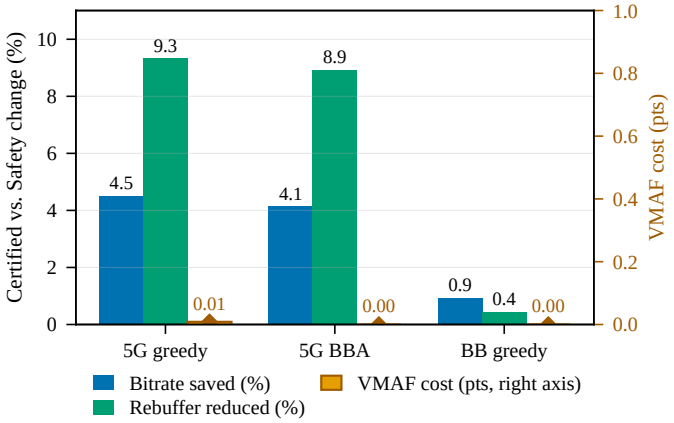


Figure 3: Certified vs. safety banking (204 episodes, ε=1.0). Bars: bitrate saved (%), rebuffering reduced (%) (left axis), and mean VMAF cost, Safety minus Certified in VMAF points, floored at 0 (right axis), for two 5G hosts and one broadband (BB) host.

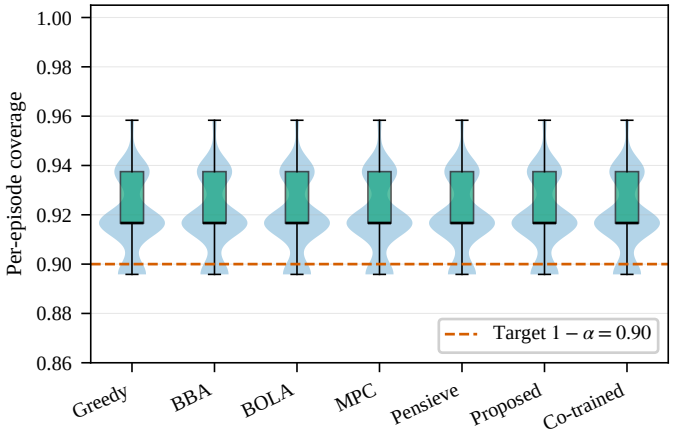


Figure 4: Per-episode conformal coverage of the certified arm (α=0.10; dashed target 0.90). The distribution is host-invariant (RQ2).

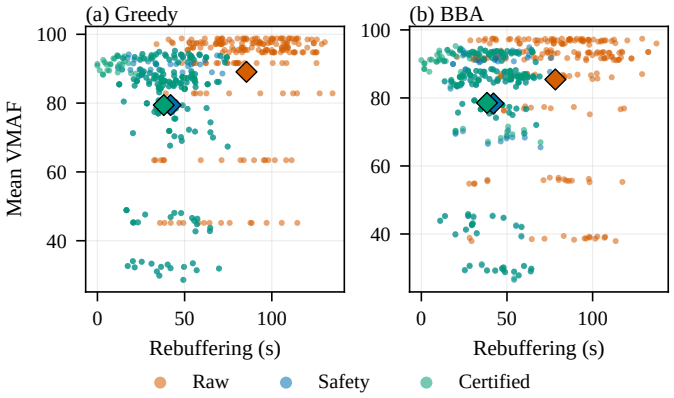


Figure 5: Per-episode operating points on paired 5G seeds. (a) Greedy. (b) BBA. Diamonds mark arm means.

At a nearly identical bitrate saving, static ε=2 therefore achieves a larger rebuffering reduction at a slightly smaller mean VMAF cost than CPS-P in this evaluation. CPS-P differs in how it spends the budget: it keeps ε=1.0 while the buffer is comfortable and widens toward $\varepsilon_{\text{risk}}$ only under forecast risk. We did not measure quality separately in stable periods, so the present data do not establish an advantage of CPS-P over a well-chosen static budget.

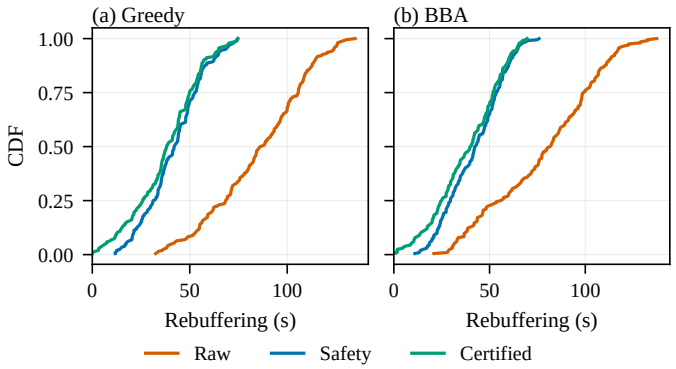


Figure 6: Empirical CDFs of episode rebuffering. (a) Greedy. (b) BBA.

Table 5: Paired arms on primary 5G test traces (204 episodes, ε=1.0, α=0.10). Reb. is mean rebuffering per episode; Cover. is empirical conformal coverage; Interv. is shield intervention rate.

Host	Arm	Reb.	VMAF	Bitrate	Cover.	Interv.
Greedy	Raw	85.5 s	89.1	6000	—	0.0%
	Safety	41.9 s	79.4	3131	0.919	65.8%
	Certified	38.0 s	79.4	2990	0.919	70.4%
BBA	Raw	78.2 s	85.4	4662	—	0.0%
	Safety	42.1 s	78.4	2809	0.919	58.1%
	Certified	38.3 s	78.5	2692	0.919	66.1%
BOLA	Raw	77.6 s	83.2	4627	—	0.0%
	Safety	42.4 s	77.5	2782	0.919	54.2%
	Certified	38.7 s	77.7	2667	0.919	61.1%
RobustMPC	Raw	71.3 s	77.4	3175	—	0.0%
	Safety	40.0 s	74.9	2377	0.919	28.3%
	Certified	36.3 s	74.9	2287	0.919	31.1%
Pensieve	Raw	0.1 s	72.2	722	—	0.0%
	Safety	0.1 s	70.8	689	0.919	7.3%
	Certified	0.1 s	70.8	689	0.919	7.4%

Certified–safety Wilcoxon: greedy bitrate p=8.5×10^{-4}, rebuffering p=8.5×10^{-4}; BBA bitrate p=4.2×10^{-16}, rebuffering p=2.5×10^{-4}. CPS-P predictive banking defaults in Table 4.

The broadband suite marks the edge of these gains. On source-disjoint real-measured broadband traces from FCC MBA and Norway/Nornet [48, 49], spanning 27 held-out traces and 204 paired video–trace episodes, greedy certified–safety bitrate savings shrink to 0.9%; see Table 6. Top rungs are seldom jointly feasible and saturated, so banking has little remaining headroom. Even so, safety vs. raw still reduces rebuffering by 98.7%. Raw greedy already stalls for most of each broadband session, as in Table 6. Because capacity limits keep saturated top rungs out of the feasible set, the conformal feasibility layer carries the main control burden. On these 27 held-out traces at α=0.10, empirical conformal coverage is 0.890, near the 0.90 target. The primary 5G pool, over 204 episodes, reaches 0.919; see Table 5 and Fig. 4. The "BB greedy" group in Fig. 3 makes the regime split explicit: banking savings are largest on the primary 5G suite. This split follows from the mechanism: CPS banking pays off precisely when the ladder knee sits within reach, and broadband places it out of reach.

Table 6: Paired greedy arms on real broadband traces (204 episodes over 27 held-out FCC/Norway traces, ε=1.0, α=0.10).

Arm	Reb.	VMAF	Bitrate	Cover.	Interv.
Raw	1216.1 s	89.08	6000	—	0.0%
Safety	16.2 s	69.58	1180	0.890	97.5%
Certified	16.2 s	69.58	1169	0.890	97.9%

Coverage under injected drift. To stress-test exchangeability without new traces, we replay the primary 5G pool with a mid-episode throughput shock on chunks 28–37: multiply trace throughput by 0.3, mimicking a sudden LOS loss or an NLOS dip. We report two related views, neither of which should be confused with the inclusive headline 0.919. First, phase-mean coverage over valid chunks after a 20-chunk conformal warm-up is 0.928 pre-shock with n=1,632, 0.929 during the shock with n=2,040, and 0.944 post-shock with n=2,040, all against the 0.90 target on the greedy certified arm over 204 episodes. Those phase averages therefore stay at or above the target through the shock. Second, Fig. 7 plots episode-averaged *rolling* coverage with window W=20 chunks under the same shock protocol. That rolling curve starts near 0.69 in the grey warm-up band because each window still mixes fallback residuals; it is a different aggregation from the inclusive per-chunk mean 0.919 and from the post–warm-up rate 0.931 on the unshocked primary pool. The sliding residual window then keeps phase-mean coverage on target after calibration, consistent with Section 4.1.

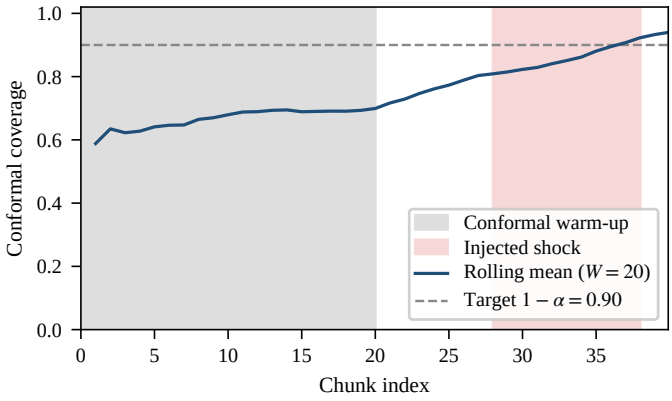


Figure 7: Rolling coverage (W=20) under a 0.3× throughput shock (red, chunks 28–37). Grey: warm-up, rolling mean near 0.69. Dashed: target 0.90. Not the inclusive coverage 0.919.

5.4. Learned hosts and co-design

We report two PPO settings separately. The primary host is a content-aware policy trained on source-disjoint broadband traces and evaluated, without further training, on 5G—the model-agnostic deployment case (Table 7). The co-design pair comprises two policies trained on matched 5G traces—one shield-at-eval-only, one co-trained with CPS in the loop—compared on their certified arms (Table 8 and Fig. 8). On the primary broadband-trained policy, certified banking vs. safety saves 35.3% bitrate and reduces rebuffering by 79.3%, with p=6.2×10^{-18}, at VMAF Δ= − 0.33 and TOST within $\pm\varepsilon$. As shown in Table 7, the bitrate cut exceeds that of greedy or BBA because the Raw mean bitrate is 1589 kb/s, not the top rung.

Table 7: Paired arms for the primary PPO host on the 5G test pool (broadband-trained policy, 204 episodes, ε=1.0, α=0.10). Certified vs. safety: −35.3% bitrate, −79.3% rebuffering.

Arm	Reb.	VMAF	Bitrate	Cover.	Interv.
Raw	3.90 s	76.42	1589	—	0.0%
Safety	2.96 s	72.99	1278	0.919	18.4%
Certified	0.61 s	72.66	827	0.919	67.2%

Table 8: Co-design on 5G test (certified arm, CPS-P at eval).

Policy	VMAF	Reb. (s)	Bitrate	Cover.
Eval-only shield	72.80	0.31	801	0.919
Co-trained (CPS)	75.73	0.86	1463	0.919
Δ (co − eval)	+2.93	+0.55	—	—

The Δ row equals the certified-arm difference in the rows above. Sign-flip test floor p=5.0×10^{-5} (VMAF) and p=5.0×10^{-5} (rebuffering), each 1/20,000, not Wilcoxon tails; bootstrap ΔVMAF CI [2.61, 3.25] over 204 episodes.

Safety already lowers it to 1278 kb/s. Certified banking then lowers it to 827 kb/s. The 35.3% saving is thus measured from this mid-ladder Safety operating point, so banking is not confined to saturated upper rungs. On more conservative hosts such as PPO it also exploits mid-ladder flats and inversions (Section 3.3) to cut rebuffering by 79.3% inside the perceptual budget. Under CPS-P, banking saves 21.9% bitrate and reduces rebuffering by 29.6%, mirroring the predictive-banking gains on non-learned hosts in Section 5.3.

Co-design. Both policies are trained on matched 5G traces with a seed-disjoint test. Under CPS-P at eval, the co-trained policy's certified arm attains VMAF 75.73 and bitrate 1463 kb/s, against 72.80/801 for shield-at-eval-only operation. The Δ row in Table 8 is the difference of those certified-arm means, +2.93 VMAF and +0.55 s rebuffering, and it matches the rows above: co-trained minus shield-at-eval-only VMAF, 75.73 − 72.80, and rebuffering, 0.86 − 0.31 s. The fidelity gain can be weighed against its stall cost through QoE(w) = $\overline{\text{VMAF}} - w \cdot \text{rebuffer(s)}$. The mean-QoE crossover is $w^\star$=5.31 VMAF points per second, the ratio of the Δ row entries: co-design favors fidelity-oriented weights $w < w^\star$; shield-at-eval-only prevails where stall aversion is stronger. Intuitively, $w^\star$ is the stall penalty at which the co-trained fidelity gain and extra rebuffering exactly balance in mean QoE; weights above $w^\star$ favor the shield-at-eval-only policy. Both reported p-values equal 5.0×10^{-5}; that value is 1/20,000, the floor of the sign-flip test, not a Wilcoxon tail probability. The bootstrap Confidence Interval (CI) [2.61, 3.25] is the magnitude statement, with stall cost [0.29, 0.85] s and crossover [3.40, 10.27]; the median crossover is 5.37, and ΔVMAF is positive in 100.0% of resamples. Co-design uses a single training seed per policy; see Section 6.

5.5. Per-chunk (non-monotone) ladder ablation (RQ4)

The headline runs apply the knee rule to each title's pooled session-mean ladder in Table 3: four of twelve ladders are fully nondecreasing, while the other eight are non-monotone only at mid-ladder from 750 to 1200 kbps. The top three rungs remain nondecreasing on every title. We therefore test whether banking remains effective on the actual per-chunk ladders, which are

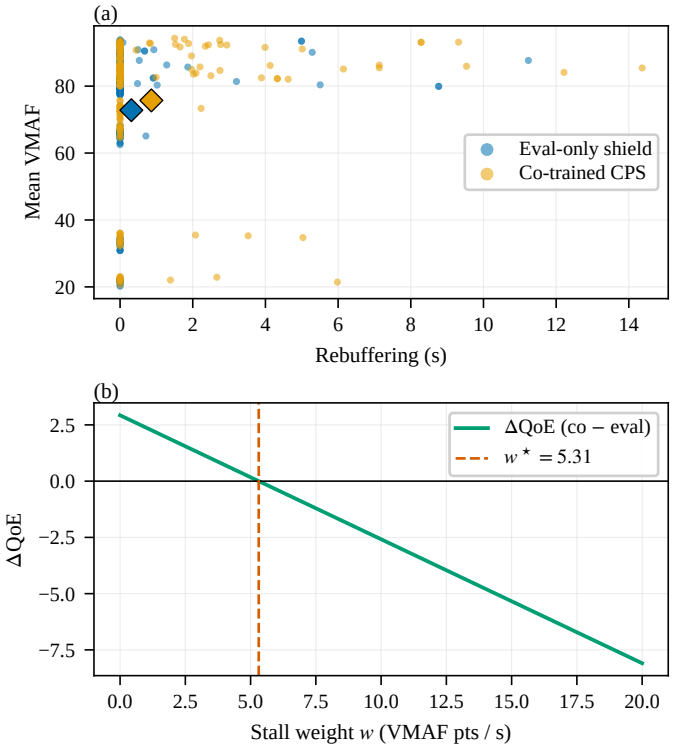


Figure 8: Co-design on the 5G test (certified arm, CPS-P). (a) VMAF–rebuffer points (diamonds: means). (b) Mean ΔQoE(w); dashed line: $w^\star$=5.31.

often non-monotone: 27.3% of the 110 chunks hold at least one inverted adjacent pair; see Fig. 2. The per-chunk ablation uses frame-level VMAF logs on Big Buck Bunny, Tears of Steel, and Sintel, the same three titles as Fig. 2. The ablation reads 110 chunks from the aggregated per-chunk export; the figure aggregates 171 chunks from raw per-frame JSON, so the two counts differ. To test this, we replay the banking stage on the per-chunk logs. We compare a shield that sees only the pooled ladder with one that sees the per-chunk ladder; see Table 9 and Fig. 9. Banking alone depends on VMAF; the feasibility projection ignores it and stays identical in both cases, which isolates the effect of the pooling simplification.

Three findings hold at the default ε=1.0, spanning correctness, savings, and pooling error. First, the knee rule remains well defined on inverted ladders and banks more, not less: mean saved bitrate is 47.0% per-chunk against 45.8% pooled, because a lower rung that inverts above the proposal satisfies the budget immediately—inversions widen banking opportunity rather than breaking the rule. Second, a per-chunk shield keeps realized per-chunk perceptual cost within budget by construction, with mean 0.33 VMAF points and never above ε, so the ablation validates chunk-wise enforcement when true ladders are available. Third, pooling is not free. Judged against the actual per-chunk VMAF, the pooled-ladder shield overruns the budget on 11.8% of chunks, with mean cost 0.34 points, which explains why session-mean evaluation can understate perceptual spend even when banking bytes look similar.

Taken together, the pooled-ladder headline numbers are, if anything, conservative with respect to banking savings. Per-chunk operation is sound: it eliminates residual budget violations

Table 9: Banking-stage ablation on real per-chunk VMAF ladders (greedy proposal, 110 chunks); the feasibility stage is identical in both shields.

ε	Mean knee rung		Bitrate saved (%)		Per-chunk cost (pts)	Budget viol. pooled (%)
	pooled	per-chunk	pooled	per-chunk	per-chunk	
0.5	4.56	4.32	22.9	28.9	0.02	9.1
1.0	4.13	3.91	45.8	47.0	0.33	11.8
2.0	3.56	3.35	59.8	61.3	1.01	19.1
4.0	2.87	2.83	70.5	70.6	1.94	7.3

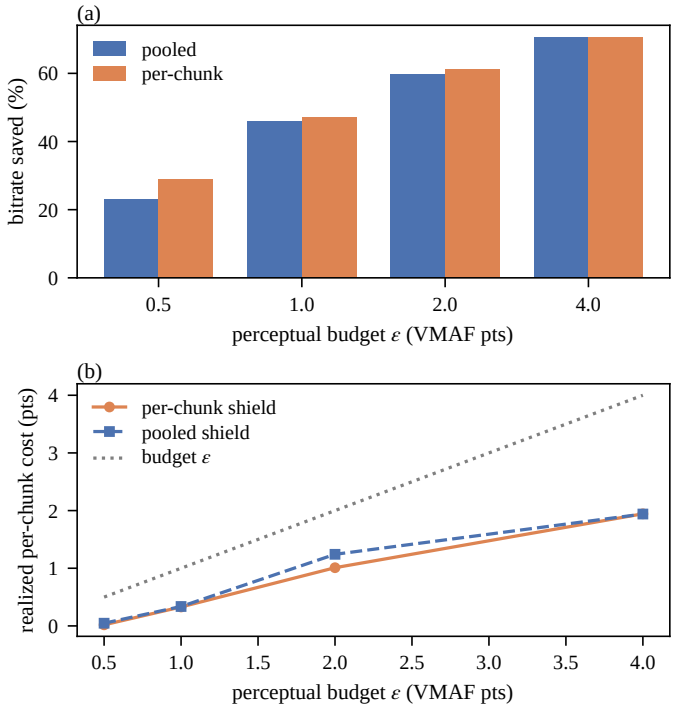


Figure 9: Per-chunk vs. pooled banking vs. ε. (a) Banked bitrate. (b) Realized per-chunk perceptual cost.

at the cost of a per-chunk VMAF feed, and it aligns the content-diversity motivation with the pooled evaluation.

5.6. *Sensitivity to ε and α (RQ4)*

The headline runs fix the pair at ε=1.0 and α=0.10. Table 10, Table 11, and Fig. 10 vary one parameter at a time on the greedy host and the primary 5G pool of 204 paired episodes over the twelve-title roster. The default cell with ε=1.0 and α=0.10 is the same evaluation as Table 5; the other cells are a one-parameter sensitivity sweep under the same seeding protocol. We first consider the perceptual budget. Sweeping $\varepsilon \in \{0.5, \ldots, 4.0\}$ at α=0.10 trades banking for fidelity. The bitrate saved relative to the Safety arm rises from 2.0% to 24.6%, while the mean VMAF change remains negligible—at most 0.48 points—and never leaves the matching $\pm\varepsilon$ envelope. Wider budgets therefore trade fidelity slack for banking headroom in a predictable way. Empirical conformal coverage remains invariant with ε. Coverage depends on the throughput trace and calibrator alone, not the banking budget, which shows that banking and the feasibility certificate operate independently. We next consider miscoverage. Sweeping $\alpha \in \{0.05, 0.10, 0.20\}$ at ε=1.0 on the same primary

Table 10: Sensitivity to the perceptual budget ε (greedy host, 5G test, α=0.10, certified vs. safety). The ε=1.0 row is the headline run of Table 5.

ε (pts)	Bitrate saved (%)	Rebuffer Δ (%)	VMAF Δ (pts)	Coverage
0.5	2.0	-3.5	+0.02	0.919
1.0	4.5	-9.3	-0.01	0.919
2.0	14.4	-30.8	-0.17	0.919
4.0	24.6	-46.1	-0.48	0.919

Table 11: Sensitivity to the conformal miscoverage α (greedy host, 5G test, ε=1.0). The α=0.10 row is the headline run of Table 5.

α	Target 1−α	Coverage	Interv. (%)	Bitrate saved (%)
0.05	0.95	0.945	73.0	4.5
0.10	0.90	0.919	70.4	4.5
0.20	0.80	0.854	35.2	6.2

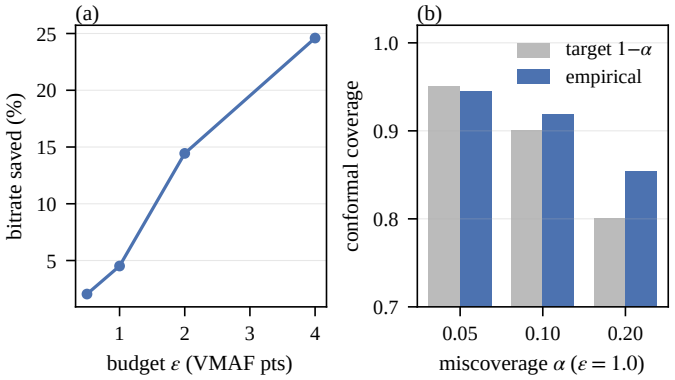


Figure 10: Sensitivity to ε and α. (a) Bitrate saved as ε grows (α=0.10). (b) Coverage vs. 1−α (ε=1.0).

5G pool moves empirical coverage in step with the finite-sample target 1−α: coverage is 0.945, 0.919, and 0.854 for targets 0.95, 0.90 (default), and 0.80, respectively. Tightening α raises both coverage and the intervention rate; loosening it lowers both, as the $(n+1)$ construction predicts. At α=0.05 the mean is 0.945, marginally below the 0.95 target over 204 paired episodes; Section 4.1 explains why short calibration windows erode finite-sample coverage. The default operating point therefore lies on a smooth, well-behaved frontier rather than a narrowly tuned optimum, with headline savings stable once ε and α are fixed.

6. Conclusion

We presented CPS, a conformal shielding layer for client-side ABR. It attaches to an existing ABR controller as a lightweight wrapper, needs no retraining, and can also wrap a learned policy. CPS combines three parts: bandwidth banking at the VMAF knee, an online conformal throughput lower bound with finite-sample coverage under residual exchangeability, and a certified buffer-feasibility test. We evaluated CPS on 204 paired 5G episodes from twelve titles with diverse ladders. Relative to the safety-only conformal projection, banking lowered delivered bitrate by 3.8–4.5% and rebuffering by 8.9–9.3%. These savings held for every non-learned host to which CPS was applied, from greedy to RobustMPC. Mean VMAF stayed inside the tolerance ±1.0 VMAF points. Empirical coverage of the throughput lower bound was 0.919, against a target of 0.90, including warm-up chunks that used the fallback scale. On the primary 5G suite, conformal safety cut mean greedy rebuffering from 85.5 s to 41.9 s per ∼192 s episode. Certified banking trimmed it further to 38.0 s. Those gains accrued where session-mean ladders saturated at the top and the host requested those rungs. Optional CPS-P and static budget sweeps (Sections 5.3 and 5.6) showed that widening ε can deepen bitrate savings. At a comparable bitrate saving, static ε=2 cut greedy rebuffering more than CPS-P (30.8% vs. 12.6%) at a slightly smaller mean VMAF cost; CPS-P's selective widening showed no measurable advantage on these metrics. Two regimes appeared. Banking supplied the savings while the ladder knee lay inside the observable operating range. Under broadband stress, the knee moved beyond that range, and the feasibility projection accounted for the stall reduction relative to raw greedy. A single-seed experiment co-trained PPO with CPS in the loop (Section 5.4). It gained +2.93 VMAF at +0.55 s of rebuffering, with a QoE crossover weight $w^\star$=5.31. Spare perceptual quality therefore becomes a budget that the client can measure and spend under a finite-sample guarantee.

Limitations and future work. Coverage holds at the 1−α level when calibration residuals are exchangeable (Section 4.1). The present evaluation used a chunk-level simulator without transport-stack or CDN dynamics [6], a synthetic 5G suite, real FCC and Norway/Nornet broadband traces [48, 49], twelve public titles [53, 54], a VMAF QoE surrogate [8], a TOST margin equal to ε=1.0 [20, 47], and one co-design training seed per policy (Table 8). Empirical coverage was 0.890 on the 27 broadband traces at α=0.10 and 0.945 vs. 0.95 on the primary 5G suite at α=0.05 (Table 11); savings scaled with ladder saturation and host aggressiveness (Table 3; Section 5.2). These bounds motivate the next steps: deployment in a production player such as `dash.js` [55], weighted conformal updates under transport-layer (TCP/QUIC) drift [43, 52], multi-seed RL co-training, packet-level or field 5G evaluation, Comyco-style learned hosts, Conditional Value-at-Risk (CVaR) objectives [56], and telemetry-driven ε and α. The feasibility check must also use the player's delayed buffer signal inside a soft real-time budget beside the ABR loop; the chunk-level micro-benchmark is ∼130 μs (median).

Data availability

The CPS implementation, chunk-level simulator, evaluation scripts, per-title session-mean VMAF ladders, and per-episode result exports underlying all tables and figures are available from the corresponding author upon reasonable request. A persistent archive with a DOI is planned for deposit upon acceptance. The FCC Measuring Broadband America and Norway/Nornet throughput traces used in the broadband suite are third-party public datasets available from their original providers [48, 49]. The synthetic 5G trace generator and configuration files used for the primary suite are available from the corresponding author on the same terms.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Saeed Zarbi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.

Leili Farzinvash: Conceptualization, Supervision, Project administration, Writing – review & editing.

Pedram Salehpour: Supervision, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude in order to improve language, clarity, and academic wording. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

[1] T.-Y. Huang, R. Johari, N. McKeown, M. Trunnell, M. Watson, A buffer-based approach to rate adaptation: Evidence from a large video streaming service, in: Proc. ACM SIGCOMM, 2014, pp. 187–198.

[2] K. Spiteri, R. Urgaonkar, R. K. Sitaraman, BOLA: Near-optimal bitrate adaptation for online videos, IEEE/ACM Trans. Netw. 28 (4) (2020) 1698–1711.

[3] X. Yin, A. Jindal, V. Sekar, B. Sinopoli, A control-theoretic approach for dynamic adaptive video streaming over HTTP, in: Proc. ACM SIGCOMM, 2015, pp. 325–338.

[4] H. Mao, M. Alizadeh, I. Menache, S. Kandula, Neural adaptive video streaming with pensieve, in: Proc. ACM SIGCOMM, 2017, pp. 197–210.

[5] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, arXiv:1707.06347 (2017).

[6] A. Narayanan, X. Zhang, R. Zhu, A. Hassan, S. Jin, X. Zhu, X. Zhang, D. Rybkin, Z. Yang, Z. M. Mao, F. Qian, Z.-L. Zhang, A variegated look at 5G in the wild: Performance, power, and QoE implications, in: Proc. ACM SIGCOMM, 2021, pp. 610–625. doi:10.1145/3452296.3472923.

[7] Cisco, Cisco annual internet report (2018–2023) white paper, https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet-report/white-paper-c11-741490.html (Mar. 2020).

[8] M. Seufert, S. Egger, M. Slanina, T. Zinner, T. Hoßfeld, P. Tran-Gia, A survey on quality of experience of HTTP adaptive streaming, IEEE Commun. Surv. Tutor. 17 (1) (2015) 469–492. doi:10.1109/COMST.2014.2360940.

[9] Z. Li, A. Aaron, I. Katsavounidis, A. Moorthy, M. Manohara, Toward a practical perceptual video quality metric, Netflix Technology Blog, https://netflixtechblog.com/toward-a-practical-perceptual-video-quality-metric-653f208b9652 (2016).

[10] M. Darwich, K. Khalil, Y. Ismail, A. Abdelgawad, M. Bayoumi, Conformal-ABR: Distribution-free risk-controlled bitrate selection for HTTP adaptive streaming, in: Proc. IEEE International Conference on Computing and Machine Intelligence (ICMI), 2026, pp. 1–6. doi:10.1109/ICMI68585.2026.11539758.

[11] H. Xie, J. Zhu, Z. Shao, C. Fan, Z. Zhang, G. Yang, P. Luo, SafeSABR: Risk-calibrated adaptive bitrate streaming over Starlink networks (2026). arXiv:2605.23560, doi:10.48550/arXiv.2605.23560.

[12] Z. Zhong, M. Liu, L. Yang, Y. Wang, Y. Xu, J.-N. Hwang, Video streaming with Kairos: An MPC-based ABR with streaming-aware throughput prediction, in: Proc. 35th Workshop on Network and Operating System Support for Digital Audio and Video (NOSSDAV), 2025, pp. 84–90. doi:10.1145/3712678.3721886.

[13] T. Huang, C. Zhou, R.-X. Zhang, C. Wu, X. Yao, L. Sun, Comyco: Quality-aware adaptive video streaming via imitation learning, in: Proc. ACM International Conference on Multimedia (MM), 2019, pp. 429–437. doi:10.1145/3343031.3351014.

[14] X. Hu, A. Ghosh, X. Liu, Z.-L. Zhang, N. Shroff, COREL: Constrained reinforcement learning for video streaming ABR algorithm design over mmWave 5G, in: Proc. IEEE International Workshop on Communications Quality and Reliability (CQR), 2023, pp. 1–6. doi:10.1109/CQR59928.2023.10317803.

[15] S. Gu, L. Yang, Y. Du, G. Chen, F. Walter, J. Wang, A. Knoll, A review of safe reinforcement learning: Methods, theories, and applications, IEEE Trans. Pattern Anal. Mach. Intell. 46 (12) (2024) 11216–11235. doi:10.1109/TPAMI.2024.3457538.

[16] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, U. Topcu, Safe reinforcement learning via shielding, Proc. AAAI Conf. Artificial Intelligence 32 (1) (2018) 2669–2678.

[17] B. Könighofer, R. Bloem, N. Jansen, S. Junges, S. Pranger, Shields for safe reinforcement learning, Commun. ACM 68 (11) (2025) 80–90. doi:10.1145/3715958.

[18] V. Vovk, A. Gammerman, G. Shafer, Algorithmic Learning in a Random World, Springer, 2005.

[19] A. N. Angelopoulos, S. Bates, Conformal prediction: A gentle introduction, Found. Trends Mach. Learn. 16 (4) (2023) 494–591. doi:10.1561/2200000101.

[20] D. Lakens, Equivalence tests: A practical primer for *t* tests, correlations, and meta-analyses, Soc. Psychol. Personal. Sci. 8 (4) (2017) 355–362. doi:10.1177/1948550617697177.

[21] Y. Sun, X. Yin, J. Jiang, V. Sekar, F. Lin, N. Wang, T. Liu, B. Sinopoli, CS2P: Improving video bitrate selection and adaptation with data-driven throughput prediction, in: Proc. ACM SIGCOMM, 2016, pp. 272–285.

[22] A. Bentaleb, A. Taani, A. C. Begen, R. Zimmermann, S. Harous, A survey on bitrate adaptation schemes for streaming media over HTTP, IEEE Commun. Surv. Tutor. 21 (1) (2019) 562–585.

[23] L. Peroni, S. Gorinsky, An end-to-end pipeline perspective on video streaming in best-effort networks: A survey and tutorial, ACM Comput. Surv. 57 (12) (2025). doi:10.1145/3742472.

[24] C. Timmerer, H. Amirpour, F. Tashtarian, S. Afzal, A. Rizk, M. Zink, H. Hellwagner, HTTP adaptive streaming: A review on current advances and future challenges, ACM Trans. Multimedia Comput. Commun. Appl. 21 (7) (2025) 198:1–198:27. doi:10.1145/3736306.

[25] R. A. K. Fezeu, E. Ramadan, W. Ye, B. Minneci, J. Xie, A. Narayanan, Z.-L. Zhang, et al., Unveiling the 5G mid-band landscape: From network deployment to performance and application QoE, in: Proc. ACM SIGCOMM, 2024, pp. 358–372. doi:10.1145/3651890.3672269.

[26] I. Sodagar, The MPEG-DASH standard for multimedia streaming over the internet, IEEE Multimed. 18 (4) (2011) 62–67. doi:10.1109/MMUL.2011.71.

[27] T. Chen, Y. Lin, N. Christianson, Z. Akhtar, S. Dharmaji, M. Hajiesmaili, A. Wierman, R. K. Sitaraman, SODA: An adaptive bitrate controller for consistent high-quality video streaming, in: Proc. ACM SIGCOMM, 2024, pp. 613–644. doi:10.1145/3651890.3672260.

[28] A. Mehrabi, M. Siekkinen, A. Ylä-Jääski, Edge computing assisted adaptive mobile video streaming, IEEE Trans. Mob. Comput. 18 (4) (2019) 787–800. doi:10.1109/TMC.2018.2850026.

[29] Z. Akhtar, Y. S. Nam, R. Govindan, S. Rao, J. Chen, E. Katz-Bassett, B. Ribeiro, Oboe: Auto-tuning video ABR algorithms to network conditions, in: Proc. ACM SIGCOMM, 2018, pp. 44–58.

[30] M. Gadaleta, F. Chiariotti, M. Rossi, A. Zanella, D-DASH: A deep Q-learning framework for DASH video streaming, IEEE Trans. Cogn. Commun. Netw. 3 (4) (2017) 703–718. doi:10.1109/TCCN.2017.2755007.

[31] M. Naresh, P. Saxena, M. Gupta, PPO-ABR: Proximal policy optimization based deep reinforcement learning for adaptive bitrate streaming, in: Proc. International Wireless Communications and Mobile Computing Conference (IWCMC), 2023, pp. 199–204. doi:10.1109/IWCMC58020.2023.10182379.

[32] R. S. Sutton, A. G. Barto, Reinforcement Learning: An Introduction, 2nd Edition, MIT Press, 2018.

[33] H. Zhang, A. Zhou, Y. Hu, C. Li, G. Wang, X. Zhang, H. Ma, L. Wu, A. Chen, C. Wu, Loki: Improving long tail performance of learning-based real-time video adaptation by fusing rule-based models, in: Proc. ACM MobiCom, 2021, pp. 775–788. doi:10.1145/3447993.3483259.

[34] F. Y. Yan, H. Ayers, C. Zhu, S. Fouladi, J. Hong, K. Zhang, P. Levis, K. Winstein, Learning in situ: A randomized experiment in video streaming, in: Proc. USENIX NSDI, 2020, pp. 495–511.

[35] E. Altman, Constrained Markov Decision Processes, CRC Press, 1999.

[36] S. Paternain, L. F. O. Chamon, M. Calvo-Fullana, A. Ribeiro, Constrained reinforcement learning has zero duality gap, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 32, 2019, pp. 7553–7563.

[37] J. Achiam, D. Held, A. Tamar, P. Abbeel, Constrained policy optimization, in: Proc. International Conf. Machine Learning (ICML), 2017, pp. 22–31.

[38] Y. Chow, O. Nachum, E. Duenez-Guzman, M. Ghavamzadeh, A Lyapunov-based approach to safe reinforcement learning, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 31, 2018, pp. 8103–8112.

[39] D. Wu, X. Wang, Y. Qiao, Z. Wang, J. Jiang, S. Cui, F. Wang, NetLLM: Adapting large language models for networking, in: Proc. ACM SIGCOMM, 2024, pp. 661–678. doi:10.1145/3651890.3672268.

[40] L. Jia, C. Zhou, C. Li, J. Liu, L. Sun, Towards user-level QoE: Large-scale practice in personalized optimization of adaptive video streaming, in: Proc. ACM SIGCOMM, 2025, pp. 1154–1166. doi:10.1145/3718958.3750526.

[41] J. Chen, Y. Yu, L. Wang, Y. Chen, T. Huang, L. Sun, Enhanced bandwidth measurement and robust rate adaptation for low-latency live streaming, in: Proc. IEEE INFOCOM, 2025, pp. 1–10. doi:10.1109/INFOCOM55648.2025.11044628.

[42] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, T. Schuster, Conformal risk control, Proc. Int. Conf. Learning Representations (ICLR) (2024).

[43] A. N. Angelopoulos, R. F. Barber, S. Bates, Online conformal prediction with decaying step sizes, in: Proc. Int. Conf. Machine Learning (ICML), Vol. 235 of PMLR, 2024, pp. 1616–1630.

[44] L. Lindemann, Y. Zhao, X. Yu, G. J. Pappas, J. V. Deshmukh, Formal verification and control with conformal prediction: Practical safety guarantees for autonomous systems, IEEE Control Syst. Mag. 45 (6) (2025) 72–122. arXiv:2409.00536.

[45] M. Kwon, T. Ingebrand, U. Topcu, L. Feng, Adaptive shielding for safe reinforcement learning under hidden-parameter dynamics shifts (2025). arXiv:2506.11033.

[46] ITU-T, Recommendation ITU-T P.910: Subjective video quality assessment methods for multimedia applications, International Telecommunication Union (2023).

[47] V. V. Menon, H. Amirpour, M. Ghanbari, C. Timmerer, Perceptually-aware per-title encoding for adaptive video streaming, in: Proc. IEEE International Conference on Multimedia and Expo (ICME), 2022, pp. 1–6. doi:10.1109/ICME52920.2022.9859744.

[48] Federal Communications Commission, Measuring broadband america: Fixed broadband reports, https://www.fcc.gov/reports-research/reports/measuring-broadband-america (2015).

[49] A. Kvalbein, D. Baltrūnas, K. Evensen, J. Xiang, A. Elmokashfi, S. Ferlin, The nornet edge platform for mobile broadband measurements, Comput. Netw. 61 (2014) 88–101.

[50] M. J. Neely, Stochastic Network Optimization with Application to Communication and Queueing Systems, Morgan & Claypool, 2010.

[51] A. Raffin, A. Hill, M. Ernestus, A. Gleave, A. Kanervisto, N. Dormann, Stable-Baselines3: Reliable reinforcement learning implementations, J. Mach. Learn. Res. 22 (268) (2021) 1–8.

[52] R. F. Barber, E. J. Candès, A. Ramdas, R. J. Tibshirani, Conformal prediction beyond exchangeability, Ann. Stat. 51 (2) (2023) 816–845. doi:10.1214/23-AOS2276.

[53] Blender Foundation, Blender studio open movies, https://studio.blender.org/films/, accessed Oct. 2026 (2026).

[54] Xiph.org Foundation, Xiph.org video test media [derf's collection], https://media.xiph.org/video/derf/, accessed Oct. 2026 (2026).

[55] K. Spiteri, R. K. Sitaraman, D. Sparacio, From theory to practice: Improving bitrate adaptation in the DASH reference player, in: Proc. ACM Multimedia Systems Conference (MMSys), 2018, pp. 123–137. doi:10.1145/3204949.3204953.

[56] R. T. Rockafellar, S. Uryasev, Optimization of conditional value-at-risk, J. Risk 2 (2000) 21–42.